

Riding the Leaders: Sector Relative-Strength Momentum

Strategy proposal for Timefolio Road to Fund Manager #13 (Oct–Nov 2026)

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I. Objective and Market

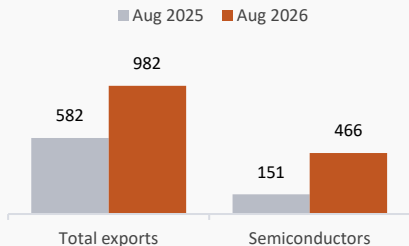
1. Objective: Return Ranking

	Return ranking (chosen)	Management skill
Evaluated on	Absolute return	Return / holding diversification, volatility
Reward	KRW 10M (1st) – 1M (5th)	Internship offer for all A+
Favored style	Concentrated, trend-following	Diversified, low-volatility

Source: DART 2026-2 Virtual Trading guideline; Timefolio contest manual

2. Market Context: Oct–Nov 2026

Korean exports, August (USD 100M)



+209% YoY

Semiconductor exports, Aug 2026 (monthly record)

8,400 → 6,000s

KOSPI: 1H peak, now range-bound

KRW 140tn → 100tn

Investor deposits since June

Strong fundamentals, flat index, wide sector dispersion. Returns depend on which sector you hold, not on market beta.

Source: DART 2026-2 guideline (MOTIE export data). Aug 2025 values back-calculated from reported growth rates.

3. Hypothesis and Evidence

Hypothesis: Leading sectors keep outperforming over a two-month horizon, and the strongest stocks within them outperform their peers.



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Study	Finding	Used in
Moskowitz & Grinblatt (1999)	Industry momentum explains a large share of stock-level momentum	Step 1: sector selection
Jegadeesh & Titman (1993)	3–12 month winners keep outperforming for 3–12 months	3M / 6M return score
George & Hwang (2004)	Stocks near their 52-week high earn higher future returns	52-week-high proximity

II. Strategy Design

4. Strategy Overview

Step	Rule	Output
Universe	KOSPI + KOSDAQ common stocks	~2,500
Filter	Market cap $\geq$ KRW 1tn, 20-day avg. trading value $\geq$ KRW 5bn, no admin / warning issues	A few hundred
1. Sectors	Top 3 by relative strength, index above 60-day MA	3 sectors
2. Stocks	Top 3 per sector by momentum and 52-week-high proximity	9 stocks
Portfolio	Weekly rebalance at Friday close, no stop-loss	9 holdings

Timing: scores computed shortly before Friday's close; orders in the closing auction.

5. Step 1: Sector Score

$$\text{Score} = 0.2 \cdot \text{RS}(1\text{M}) + 0.4 \cdot \text{RS}(3\text{M}) + 0.4 \cdot \text{RS}(6\text{M})$$

$\text{RS}(k)$ = sector index return over k months – KOSPI return over k months

Selection. Hold the top 3 sectors by score.

Trend filter. A sector is eligible only while its index is above its 60-day moving average.

Buffer. A held sector is replaced only when it falls out of the top 4, to limit churn.

Weights. Medium-term returns get 80% of the weight; 1-month return alone is too noisy.

Sector definition. GICS industry groups in the contest. In the backtest, 25 GICS-like groups mapped from the Korean Standard Industrial Classification (KSIC), using today's classification for the whole period.

6. Step 2: Stock Selection

Score = rank(R3M) + rank(R6M) + rank(Price / 52-week high)

Returns exclude the most recent week

Eligibility. Market cap $\geq$ KRW 1tn, 20-day avg. trading value $\geq$ KRW 5bn, no admin / warning issues, listed $\geq$ 6 trading days.

Skip last week. Removes short-term reversal noise after sharp moves.

52-week-high proximity. Investors anchor on the prior high; stocks breaking through it tend to keep trending.

Output. Top 3 stocks in each selected sector, 9 in total. A held stock stays while it ranks in the sector's top 5.

7. Portfolio and Contest Rules

Sector weights 40 / 35 / 25% by score rank, equal-weighted within each sector: 13.3 / 11.7 / 8.3% per stock.

Samsung Electronics is held at 25% if selected (contest cap 40%). Kept holdings are resized only if they drift more than 3%p from target.

Contest rule	How the strategy complies
Single stock $\leq 15\%$ (Samsung $\leq 40\%$)	Max 13.3%; Samsung max 25%
GICS sector $\leq 2 \times$ market weight	Checked before each order; excess moves to the next-ranked sector
Stocks < KRW 1tn cap $\leq 30\%$ combined	Excluded from the universe
No buys: trading value $\leq$ KRW 3bn, cap < KRW 100bn, warnings	Filters are stricter than the contest limits
Weekly turnover $\geq 5\%$	Weekly rebalancing; top-up trades in quiet weeks (23% of backtest weeks were below 5%)

Source: Timefolio Road to Fund Manager manual (investment universe, portfolio guidelines)

8. Execution and Risk Rules

Friday, before the close. Update prices and compute sector and stock scores (one script run).

Closing auction. Trade to target weights; time-sliced or closing-auction orders to limit slippage.

Turnover control. Sector replaced only when out of the top 4; stock kept while in its sector's top 5; weights resized only on drift > 3%p.

Turnover rule. If a week's turnover is below 5%, add small rebalancing trades before the close.

Risk control	First version	Final version
Stop-loss	12% trailing stop from peak	None (stops hurt in all 9 tests)
Execution	Monday close	Friday close
Downside limits	Stop + 60-day MA sector filter	60-day MA sector filter, 15% stock cap

III. Validation

9. Backtest Design

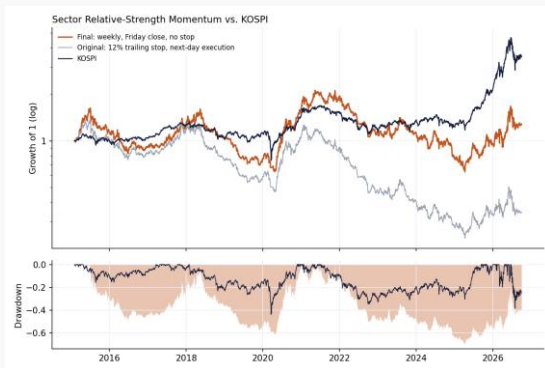
Item	Setting
Period	Feb 2015 – Sep 2026, weekly signals
Data	KRX via pykrx: adjusted daily prices, weekly market-cap snapshots. About 60% of delisted stocks are missing (no sector data), so some survivorship bias remains.
Sectors	25 GICS-like groups (KSIC mapped, today's classification applied to the past); sector return = market-cap-weighted return of member stocks
Costs	10 bp commission each way, 20 bp sell tax, 10 bp slippage each way
Benchmark	KOSPI
Metrics	CAGR, volatility, Sharpe, max drawdown, turnover; Oct–Nov returns by year

Overfitting check

36 parameter sets (3 lookback weightings $\times$ 2 / 3 / 4 sectors $\times$ 8 / 12 / 15% / no stop); all results reported.

Deflated Sharpe Ratio (Bailey & López de Prado, 2014) adjusts for the number of trials and non-normal returns.

10. Backtest Results



	Final	Original	KOSPI
CAGR	2.1%	-8.7%	11.3%
Sharpe	0.23	-0.17	0.60
Max DD	-70%	-83%	-44%
Turnover/ wk	24%	33%	—

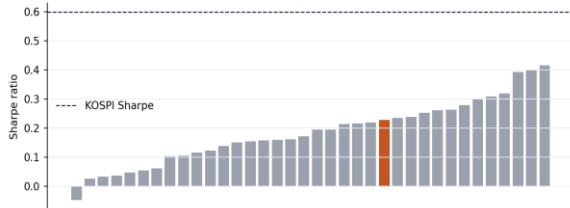
Feb 2015 – Sep 2026, after costs. Final = weekly, Friday close, no stop.

The final version trails the KOSPI by 9%p a year with a -70% drawdown.

It kept up with the market in 2020–21 and in the 2025–26 semiconductor rally, but lost ground whenever leading sectors reversed (2018–19, 2022–24). In 2025–26 the index was driven by two stocks, Samsung Electronics and SK hynix, which a 9-stock sector portfolio only partly held. The final rules were chosen after seeing the original version's results (data snooping).

11. Robustness and Contest-Period Returns

Sharpe ratio across 36 parameter sets (orange = final version)



Parameter grid

None of the 36 sets beats the KOSPI's Sharpe of 0.60 (best 0.42).

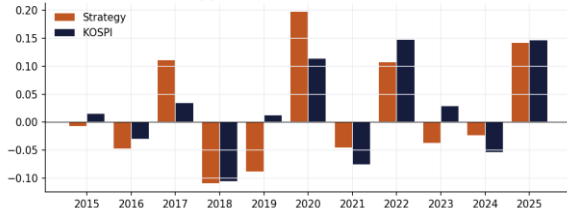
Deflated Sharpe

0.49 for the final version, 0.73 for the best set: not significant after correcting for 36 trials.

October–November

Beat the KOSPI in 4 of 11 years; returns ranged from -11% to +20%. A high-variance bet, as intended for the return ranking.

October–November return by year



Questions?

Code and backtest results: [DART-KNU / 2026-2-archive / 2026-2-quant-strategy](#)